Dsa Series Day10 Vamsi Journey

Pattern printing

Hint:

- `print("*", end="")` → prints stars in the **same row**
- `print()` → moves to the **next line** after one row is complete

⭐ 1. Solid Square Pattern

```
*****
*****
*****
*****
*****
```

```
for i in range(5):
    for j in range(5):
        print("*", end="")
    print()
```

⭐ 2. Left Half Pyramid (Stars)

```
*
**
***
```

```
****
*****
```

```
for i in range(1, 6):
    for j in range(1, i+1):
        print("*", end="")
    print()
```

⭐ 3. Left Half Pyramid (Same Number)

```
1
22
333
4444
55555
```

```
for i in range(1, 6):
    for j in range(1, i+1):
        print(i, end="")
    print()
```

⭐ 4. Left Half Pyramid (Increasing Numbers)

```
1
12
123
1234
12345
```

```
for i in range(1, 6):
    for j in range(1, i+1):
        print(j, end="")
    print()
```

⭐ 5. Inverted Left Half Pyramid (Stars)

```
*****
****
***
**
*
```

```
row = 5
for i in range(1, row+1):
    for j in range(1, row-i+1):
        print("*", end="")
    print()
```

⭐ 6. Inverted Left Half Pyramid (Same Number)

```
11111
2222
333
44
5
```

```
row = 5
for i in range(1, row+1):
    for j in range(1, row-i+1):
        print(i, end="")
    print()
```

⭐ 7. Inverted Left Half Pyramid (Increasing Numbers)

```
12345
1234
123
12
1
```

```
row = 5
for i in range(1, row+1):
    for j in range(1, row-i+1):
        print(j, end="")
    print()
```

⭐ 8. Full Pyramid (Stars)

```
    *
   ***
  *****
 *******
*********
```

```
n = 5
for i in range(n):
    for j in range(n-i-1):
        print(" ", end="")
    for j in range(2*i+1):
        print("*", end="")
    print()
```

⭐ 9. Inverted Full Pyramid

```
*********
 *******
  *****
   ***
    *
```

```
n = 5
for i in range(n):
    for j in range(i):
        print(" ", end="")
    for j in range(2*(n-i)-1):
        print("*", end="")
    print()
```

⭐ 10. Diamond Pattern

```
    *
   ***
  *****
 *******
*********
```

```
 *******
  *****
   ***
    *
```

```
n = 5
for i in range(n):
    for j in range(n-i-1):
        print(" ", end="")
    for j in range(2*i+1):
        print("*", end="")
    print()

for i in range(n-2, -1, -1):
    for j in range(n-i-1):
        print(" ", end="")
    for j in range(2*i+1):
        print("*", end="")
    print()
```

⭐ 11. Simple Increasing Stars (Single Loop Logic)

```
*
**
***
****
*****
****
***
**
*
```

```
n = 9
mid = n // 2

for i in range(n):
    if i <= mid:
        stars = i + 1
    else:
        stars = n - i
    for j in range(stars):
        print("*", end="")
    print()
```

🔹 12. Hollow Square Pattern

```
*****
*   *
*   *
*   *
*****
```

```
n = 5
for i in range(n):
    for j in range(n):
        if i == 0 or i == n-1 or j == 0 or j == n-1:
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

🔹 13. Hollow Right Triangle

```
*
**
* *
*  *
*****
```

```
n = 5
for i in range(1, n+1):
    for j in range(1, i+1):
        if j == 1 or j == i or i == n:
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

🔹 14. Number Triangle (Continuous Numbers)

```
1
2 3
4 5 6
7 8 9 10
```

```
n = 4
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
```

```
        num += 1
    print()
```

🔹 15. Floyd’s Triangle

```
1
2 3
4 5 6
7 8 9 10
```

(Same as above, but name matters for interviews)

```
n = 4
num = 1
for i in range(1, n+1):
    for j in range(i):
        print(num, end=" ")
        num += 1
    print()
```

🔹 16. Binary Triangle

```
1
01
101
0101
```

```
n = 4
for i in range(1, n+1):
    for j in range(1, i+1):
        if (i + j) % 2 == 0:
```

```
            print(1, end="")
        else:
            print(0, end="")
    print()
```

🔹 17. Pascal’s Triangle (Basic Version)

```
1
1 1
1 2 1
1 3 3 1
```

```
n = 4
for i in range(n):
    val = 1
    for j in range(i+1):
        print(val, end=" ")
        val = val * (i - j) // (j + 1)
    print()
```

🔹 18. Hollow Pyramid

```
    *
   * *
  *   *
 *     *
*********
```

```
n = 5
for i in range(n):
    for j in range(n-i-1):
```

```
        print(" ", end="")
    for j in range(2*i+1):
        if j == 0 or j == 2*i or i == n-1:
            print("*", end="")
        else:
            print(" ", end="")
    print()
```

🔹 19. Hollow Diamond

```
   *
  * *
 *   *
  * *
   *
```

```
n = 3
for i in range(n):
    for j in range(n-i-1):
        print(" ", end="")
    for j in range(2*i+1):
        if j == 0 or j == 2*i:
            print("*", end="")
        else:
            print(" ", end="")
    print()

for i in range(n-2, -1, -1):
    for j in range(n-i-1):
        print(" ", end="")
    for j in range(2*i+1):
        if j == 0 or j == 2*i:
            print("*", end="")
```

```
        else:
            print(" ", end="")
    print()
```

🔹 20. Character Triangle

```
A
A B
A B C
A B C D
```

```
n = 4
for i in range(1, n+1):
    ch = ord('A')
    for j in range(i):
        print(chr(ch), end=" ")
        ch += 1
    print()
```

🔹 21. Same Number Pyramid

```
1
22
333
4444
```

```
n = 4
for i in range(1, n+1):
    for j in range(i):
```

```
        print(i, end="")
    print()
```